

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. Examination (CL)

March-April 2018

CL 213 Building Constructions

Date: 06.04.2018, Friday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 (a) Do as directed.

[07]

- 1) The angular steps used for changing direction of the stairs, are called
(A) round steps (B) angular steps (C) winders (D) radial steps
- 2) Draw a figure showing hip roof.
- 3) In painting, the defect caused due to formation of dull patches on finished polished surface is known as _____.
- 4) Define Buttress.
- 5) Draw the figure of Squint Brick.
- 6) When immersed in water for 1 hour the 1st Class Bricks should not absorb more than _____ of their weight.
- 7) Sketch Any one.
(1) dog-legged stair (2) Open-well stair (3) Quarter- turn stair.

Q - 2 (a) Write requirements of good stair. Sketch a bifurcated stair.

[07]

(b) Explain following methods with neat diagrams :(Any Two)

[07]

- 1) Pit method 2) Single flying shore 3) Double scaffolding

OR

Q - 2 (a) Define following terms with sketch.

[07]

Baluster, Newel post, Hand rail, Winders, Landing, soffit, Head room.

(b) Defects observed in Painting.

[07]

Q - 3 (a) Explain the purpose for which sub-soil exploration is done. How do you decide the depth of exploration?

[07]

(b) Write advantages and disadvantages of precast concrete piles

[07]

OR

- Q - 3 (a)** Explain Methods of three coat lime plaster. **[07]**
- (b)** Define following terminology used in plastering work. **[07]**
Back ground, Dots, Gauging, Dubbing coat, Hacking, Keys

SECTION - II

- Q - 4 (i)** Lists commonly used roof covering materials for pitched roof. Explain any one. **[07]**
- (ii)** List types of Window. Give the recommended dimension of windows as per IS.
- Q - 5 (a)** Enlist various types of joints in stone masonry. Explain table joint in detail with neat sketches. **[07]**
- (b)** Classify the brick masonry and describe any one masonry in detail with Sketch. **[07]**

OR

- Q - 5 (a)** Draw a detailed figure of solid ground floor and enlist the factors affecting the selection of flooring material **[07]**
- (b)** Draw a figure of steel roof truss indicating all the elements. Enlist four advantages of steel roof truss over timber roof truss. **[07]**
- Q - 6 (a) Write Any Two.** **[14]**
- (1)** Discuss casement window with sketch.
- (2)** Explain collapsible steel door with sketch. Discuss construction suitability of the same.
- (3)** Draw neat sketch of following
(i) Fanlight ventilator (ii) Dormer or Gable window
- (4)** Enlist various pitched roof. Explain any one type with labeled sketch.
